Satellite Verification of India's Vibrant Villages Programme: A Compositing-Window Robustness Assessment of Border-Village Development

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Abstract

The Vibrant Villages Programme (VVP-I) is the Government of India's flagship border-area development scheme, sanctioning approximately ₹4,800 crore across 2,967 villages in five Himalayan border states/union territories since 2022–23, with 662 designated "priority" villages for Phase 1 development. A government reply on parliamentary record states that no impact assessment has ever been carried out for this programme. This study addresses that gap directly, independently testing whether VVP-I priority villages show measurable physical development using Sentinel-2 built-up-index (NDBI) and VIIRS night-lights data at 258 individually geocoded villages across three core states (Arunachal Pradesh, Sikkim, Uttarakhand) plus a small illustrative sample from a fourth (Himachal Pradesh), comparing a pre-sanction baseline (2021) against the present (2025) under two independent compositing windows. A full-year composite showed no significant increase in built-up area (Wilcoxon signed-rank, $p = 1.000$); a season-matched (June–September) composite on the same villages showed the opposite result ($p < 0.000001$) — a reversal traced to snow-cover contamination in the full-year window and monsoon cloud cover eliminating Sikkim's data entirely in the summer window. VIIRS night-lights, tested identically under both windows, showed no significant increase in either ($p = 0.050$ full-year; $p = 0.9999$ summer-matched), making it the more temporally stable — and more trustworthy — proxy. Descriptively — an illustrative two-state pattern, not a formal statistical test — Arunachal Pradesh's sanctioned budget (₹2,749.74 crore) was roughly ten times Uttarakhand's (₹270.58 crore), yet the two states' mean measured built-up-area change was nearly identical (+0.0284 vs. +0.0293), consistent with budget scale not translating proportionally into physical change. Border-proximity effects (H3) showed a stable null for built-up change ($\rho = 0.043$–0.037, both non-significant) and an unstable result for night-lights (significant only in the full-year window). Rather than resolving this instability by selectively citing the more favourable result, this study reports the instability itself as the central finding: VVP-I's claimed development is not currently demonstrable with confidence from satellite evidence, precisely the accountability gap the programme's own parliamentary record identifies.

Keywords: Vibrant Villages Programme, border development, securitization theory, satellite verification, NDBI, VIIRS night-lights, remote sensing, robustness testing

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1. Introduction

India's border villages have historically been framed in policy discourse as security liabilities — remote, under-populated, and vulnerable to depopulation toward a contested frontier. The Vibrant Villages Programme (VVP-I), approved by the Union Cabinet in February 2023, reframes them instead as development priorities, sanctioning infrastructure and livelihood investment across 2,967 villages in Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh, and Ladakh. What the programme has not received, in more than two years since sanction, is an independent, evidence-based accounting of whether the development it promised has actually occurred on the ground. Asked directly in Parliament whether any impact assessment has been carried out for VVP, the Ministry of Home Affairs answered without qualification: "No impact assessment has been carried out" (Lok Sabha Unstarred Question No. 508, 3 February 2026).

This study addresses that gap directly, using multi-temporal satellite imagery — built-up area indices and night-time light radiance — to test, village by village, whether physical development detectable from space has occurred since programme sanction, whether its magnitude tracks each state's sanctioned budget, and whether its distribution is better explained by border proximity than by developmental need, consistent with securitization theory's prediction that border infrastructure functions as a geopolitical signal as much as a welfare intervention.

Specifically, this study tests four research questions: (RQ1) whether priority villages show statistically detectable built-up area growth since programme sanction; (RQ2) whether the magnitude of that change correlates with each state's sanctioned budget; (RQ3) whether night-time light trends corroborate the built-up area findings; and (RQ4) whether development is spatially concentrated by border proximity rather than developmental need. Correspondingly, three hypotheses are tested: **H1** predicts a statistically significant post-sanction increase in built-up area; **H2** predicts that the size of this change will not be uniformly proportional to sanctioned budget, with some high-budget states showing comparatively muted physical change; and **H3** predicts that villages nearer the border/LAC will show disproportionately faster change than villages farther back within the same state, consistent with securitization theory's prediction that border proximity — not developmental need — drives priority.

2. Literature Review

2.1 Securitization Theory and Border Development

Securitization theory (Buzan, Wæver, & de Wilde, 1998) argues that framing an issue as a matter of security — rather than ordinary politics — justifies extraordinary measures and resource allocation that would not otherwise be politically available. Border-area development programmes sit naturally within this frame: investment is justified simultaneously as civilian welfare and as strategic infrastructure along a contested frontier, and the two justifications are rarely disentangled in either policy design or public reporting. This study treats VVP-I as a direct empirical test case for that theory — specifically, whether the spatial distribution of measurable development is better predicted by border proximity than by conventional developmental indicators such as population or existing infrastructure gaps.

2.2 The Vibrant Villages Programme: Policy Context and the Accountability Gap

VVP-I's sanctioned scope is documented across multiple government sources: a Rajya Sabha Unstarred Question reply (Question No. 2321, Ministry of Home Affairs, 9 August 2023) provides Sikkim's full village-wise annexure; Lok Sabha Question No. 2104 (2023) and Rajya Sabha Question No. 401 (2025) confirm Himachal Pradesh's 75 priority villages without a name-level annexure; and Lok Sabha Question No. 4360 (2025) confirms Ladakh's 35 sanctioned villages, again without a published village list. Most directly relevant to this study's premise, Lok Sabha Unstarred Question No. 508 (3 February 2026) asked the Ministry of Home Affairs specifically whether VVP's impact had been assessed; the Ministry's answer was unqualified: "No impact assessment has been carried out." No government source identified in the course of this study's data-acquisition process provides an independent, satellite-verified accounting of physical progress against this sanctioned scope — the absence this study is designed to address.

2.3 Remote Sensing Approaches to Built-Up Area and Economic Activity Detection

The Normalized Difference Built-up Index (NDBI), computed from short-wave infrared and near-infrared reflectance, is an established proxy for built-up surface extent in multi-temporal satellite comparison (Zha, Gao, & Ni, 2003). VIIRS Day/Night Band night-time radiance is a complementary, independent proxy for economic and electrification activity, with documented advantages over older DMSP-OLS night-lights products in dynamic range and spatial resolution (Elvidge, Baugh, Zhizhin, Hsu, & Ghosh, 2017). This study uses both indices in parallel specifically because they are subject to different confounds — NDBI to vegetation phenology and snow cover, VIIRS to cloud cover and sensor saturation at very low radiance — so that agreement between the two carries more evidentiary weight than either alone.

2.4 Compositing-Window Sensitivity in Multi-Temporal Satellite Analysis

Seasonal compositing choices are known to materially affect spectral-index-based change detection in high-relief terrain, where snow cover and cloud frequency both vary systematically by season and elevation. This study treats compositing-window sensitivity not as a nuisance parameter to be tuned away, but as a testable property of the result itself: a finding that reverses direction between two independently defensible compositing windows is, on its own terms, evidence that a single-window result should not be reported as confirmatory.

3. Data and Methodology

3.1 Study Design

This study covers VVP-I priority villages across five Himalayan border states/union territories, restricting the core statistical sample to the three states with complete, officially-sourced village-wise identification — Arunachal Pradesh, Sikkim, and Uttarakhand (251 villages) — while carrying Himachal Pradesh's 7 confirmed villages as a separately labelled illustrative case study and excluding Ladakh from village-level analysis entirely, pending resolution of a documented data-availability gap.

3.2 Data Sources

Variable	**Source**	**Temporal Coverage**
Village identification	State VVP-I portals; Rajya Sabha/Lok Sabha Q&A annexures	2023–2025
Village coordinates	OpenStreetMap Nominatim; ISRO Bhuvan Village Geocoding API	Current
Built-up index (NDBI)	Sentinel-2 SR Harmonized	2021, 2025
Night-lights radiance	VIIRS DNB monthly composites	2021, 2025
Border/LAC geometry	Natural Earth 10m Admin-0 Boundary Lines	Current
Budget/project counts	State-wise VVP-I sanction figures (parliamentary record)	2023

3.3 Village-Level Dataset Compilation and Geocoding

Village lists were cross-verified against officially reported aggregate counts before acceptance, then geocoded via Nominatim (primary) with Bhuvan used as a Census-linked fallback, subject to manual district validation after a direct test confirmed Bhuvan does not filter by state (a query for "Kharman," an Arunachal Pradesh village, initially returned an unrelated same-named village in Haryana). Of 559 villages attempted across the four named states, 258 were successfully geocoded (Arunachal Pradesh 186/455, Sikkim 31/46, Uttarakhand 34/51, Himachal Pradesh 7/7).

3.4 Satellite Change Detection

For each geocoded village, a 500m buffer was used to extract mean NDBI and VIIRS night-lights radiance via Google Earth Engine, comparing a 2021 baseline against 2025.

NDBI was computed using Sentinel-2 Level-2A Surface Reflectance Harmonized bands B11 (SWIR1) and B8 (NIR), with cloud masking applied via the QA60 band. Night-lights radiance was drawn from the VIIRS Day/Night Band monthly composite product (NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG).

3.5 Compositing-Window Robustness Testing

Both metrics were computed under two independent compositing windows — a full calendar year, and a season-matched June–September window — to isolate genuine change from seasonal artifacts (snow cover in the full-year window; monsoon cloud cover in the summer window), with any village lacking a valid composite in either window marked null rather than defaulted to zero.

3.6 Border-Proximity Analysis

Distance from each village to the nearest segment of Natural Earth's Admin-0 boundary line was computed via GeoPandas (UTM 44N reprojection, nearest-point distance), restricted to India-relevant boundary segments. This distance is explicitly a cartographic approximation: India's boundary with China in this region, commonly referenced as the Line of Actual Control, is disputed and has no single internationally agreed alignment, so all distance figures should be read as relative and comparative rather than as an authoritative statement of the boundary's legal position.

3.7 Statistical Testing

Before/after change was tested using the Wilcoxon signed-rank test (Wilcoxon, 1945), a paired non-parametric test appropriate given the sample's non-normal distribution. Budget-correlation (RQ2, descriptive only given n=2 states with sufficient valid data) and border-proximity correlation (H3) used Spearman's rank correlation (Spearman, 1904), robust to non-linear monotonic relationships.

4. Results

4.1 Village Coverage and Sample Composition

The geocoding pipeline resolved 258 of 559 attempted villages. Inspection of unmatched Arunachal Pradesh entries showed many are not civilian revenue villages but Border Roads Task Force camps, army staging huts, and labour-camp designations — inhabited points along the frontier absent from every available civilian geospatial database, a direct illustration of what "village-level development" consists of on a securitized border rather than merely a data gap.

4.2 Built-Up Area Change: A Compositing-Window-Sensitive Result

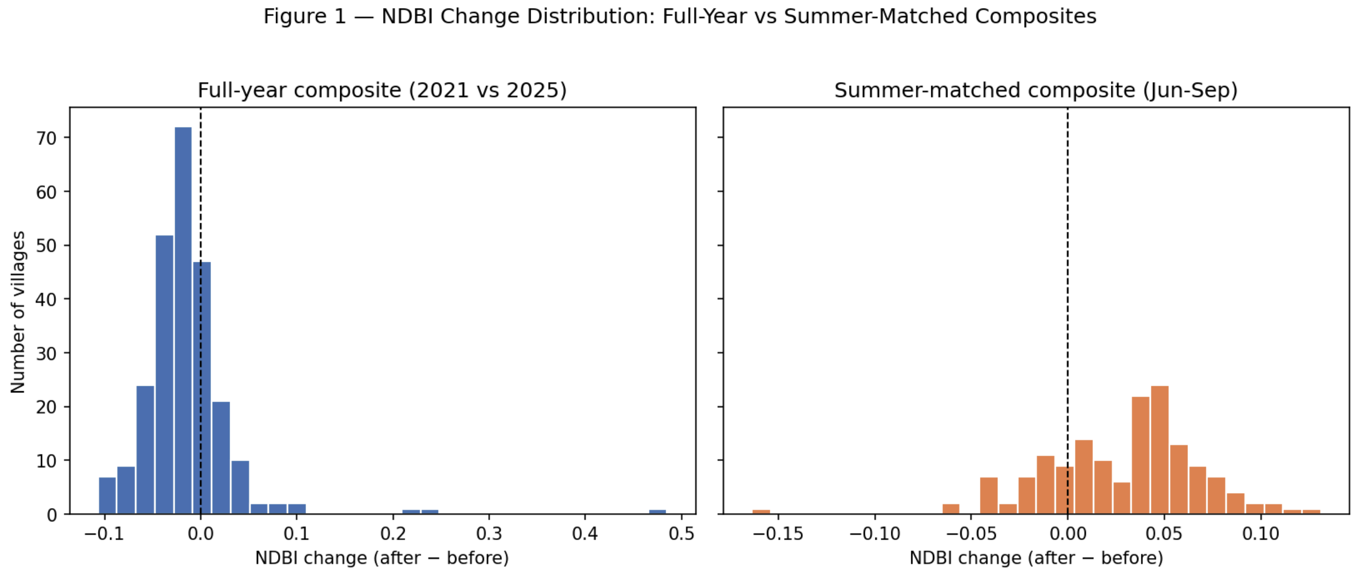

Figure 1. Distribution of NDBI change across all geocoded villages, full-year and summer-matched composites.

Under the full-year composite, mean NDBI change across the 251-village core sample showed no significant increase (Wilcoxon signed-rank, $p = 1.000$), trending slightly negative at the median. Under the season-matched (June–September) composite, the same test on the same villages ($n = 154$ with valid data after seasonal dropout) showed a highly significant increase ($p < 0.000001$) — the opposite conclusion. This reversal is attributed to snow-cover contamination varying between the 2021 and 2025 full-year composites at high altitude, a confound the summer window was designed to avoid; the summer window in turn eliminated Sikkim's data entirely, since all 31 of its geocoded villages returned zero cloud-free imagery in at least one period of that window — Sikkim sits in the peak-monsoon zone during exactly the months chosen to avoid snow.

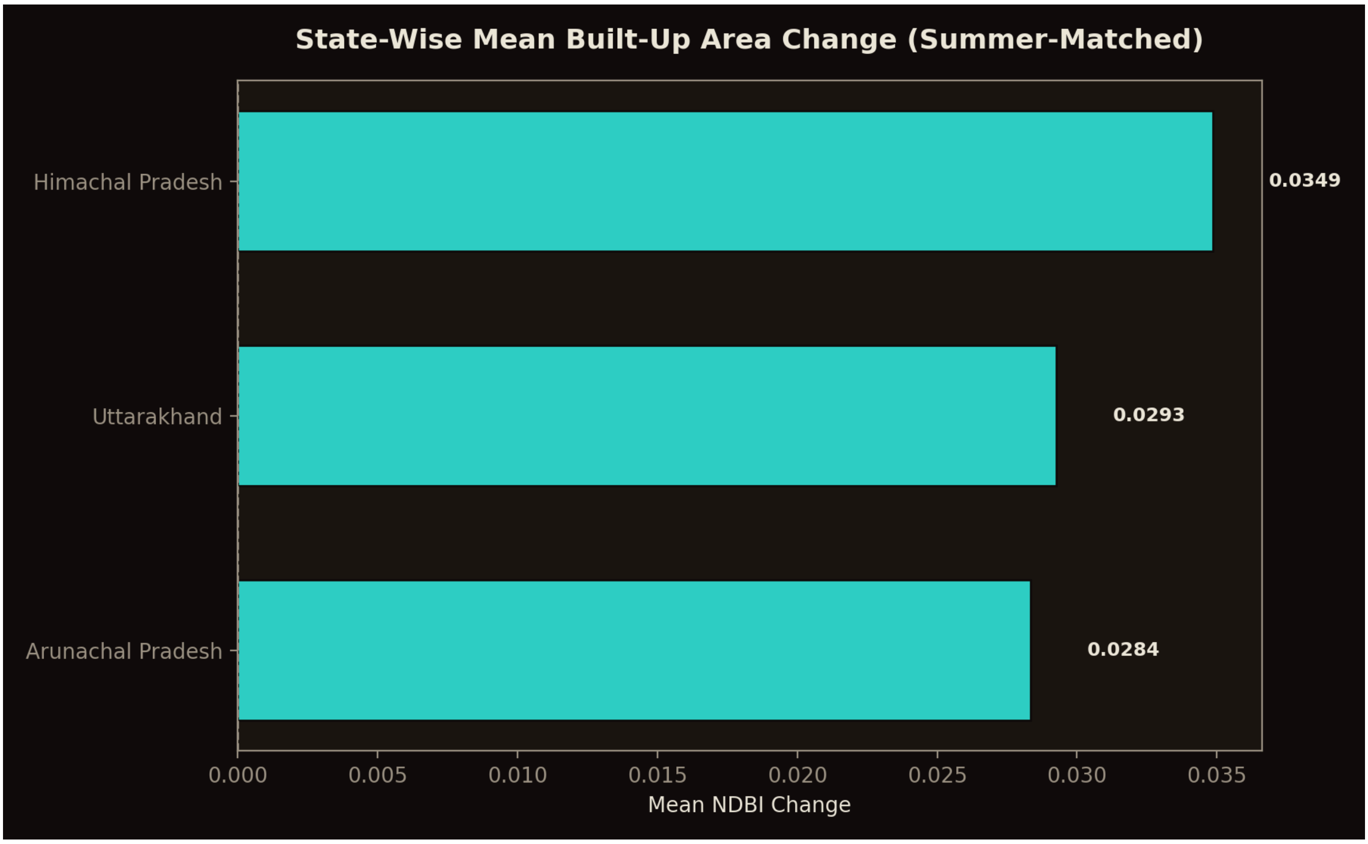

Figure 2. State-wise mean built-up-area change (summer-matched composite), by state.

4.3 Night-Lights Change: A Stable Null

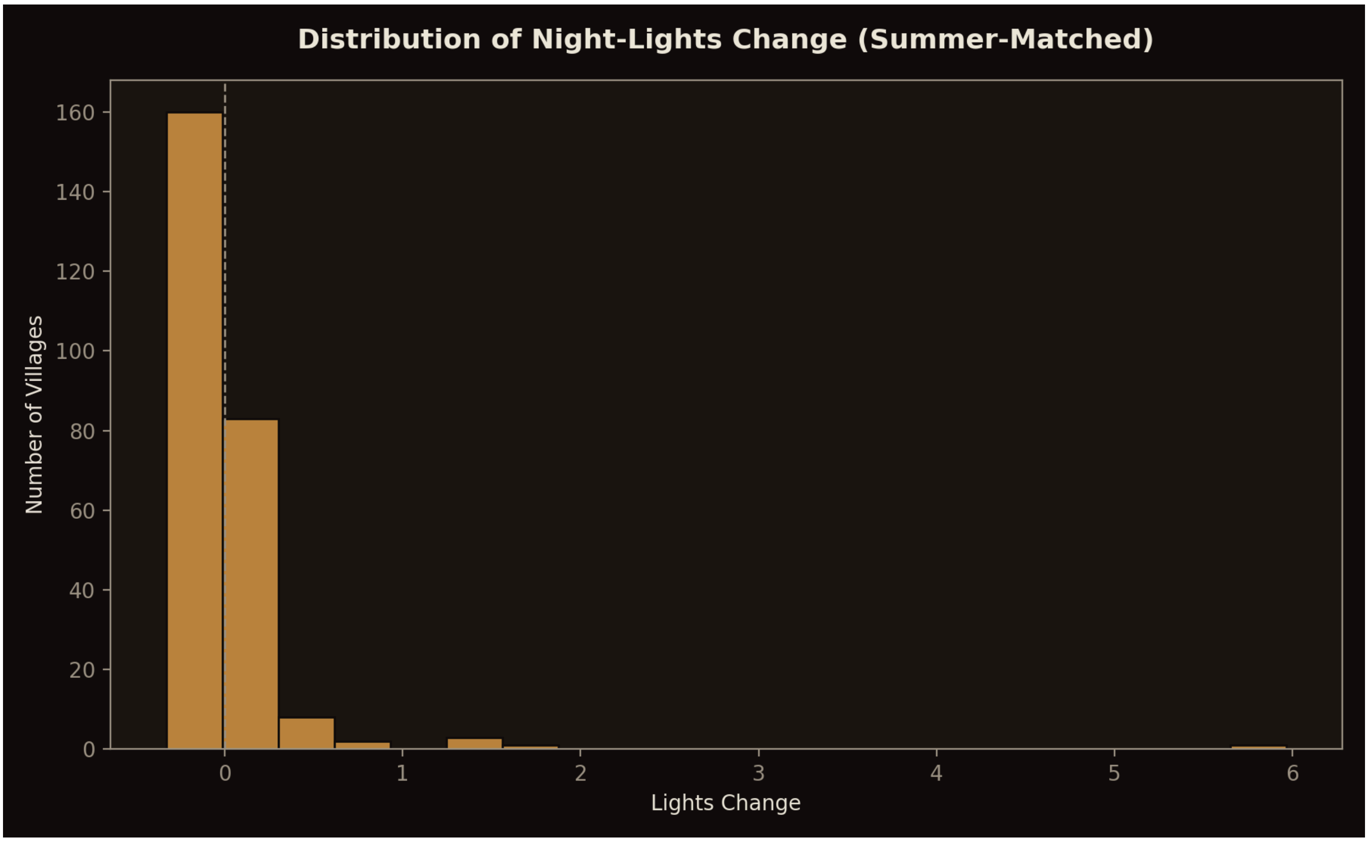

Figure 3. Distribution of VIIRS night-lights change (summer-matched composite).

VIIRS night-lights, tested identically under both compositing windows, showed no significant increase in either (full-year $p = 0.050$, borderline; summer-matched $p = 0.9999$). Because night-lights radiance is not subject to the same vegetation/snow phenology confound as a spectral built-up index, its consistency across both windows is read as the more trustworthy signal of the two — and that signal does not support a confirmed increase in night-time economic activity following programme sanction.

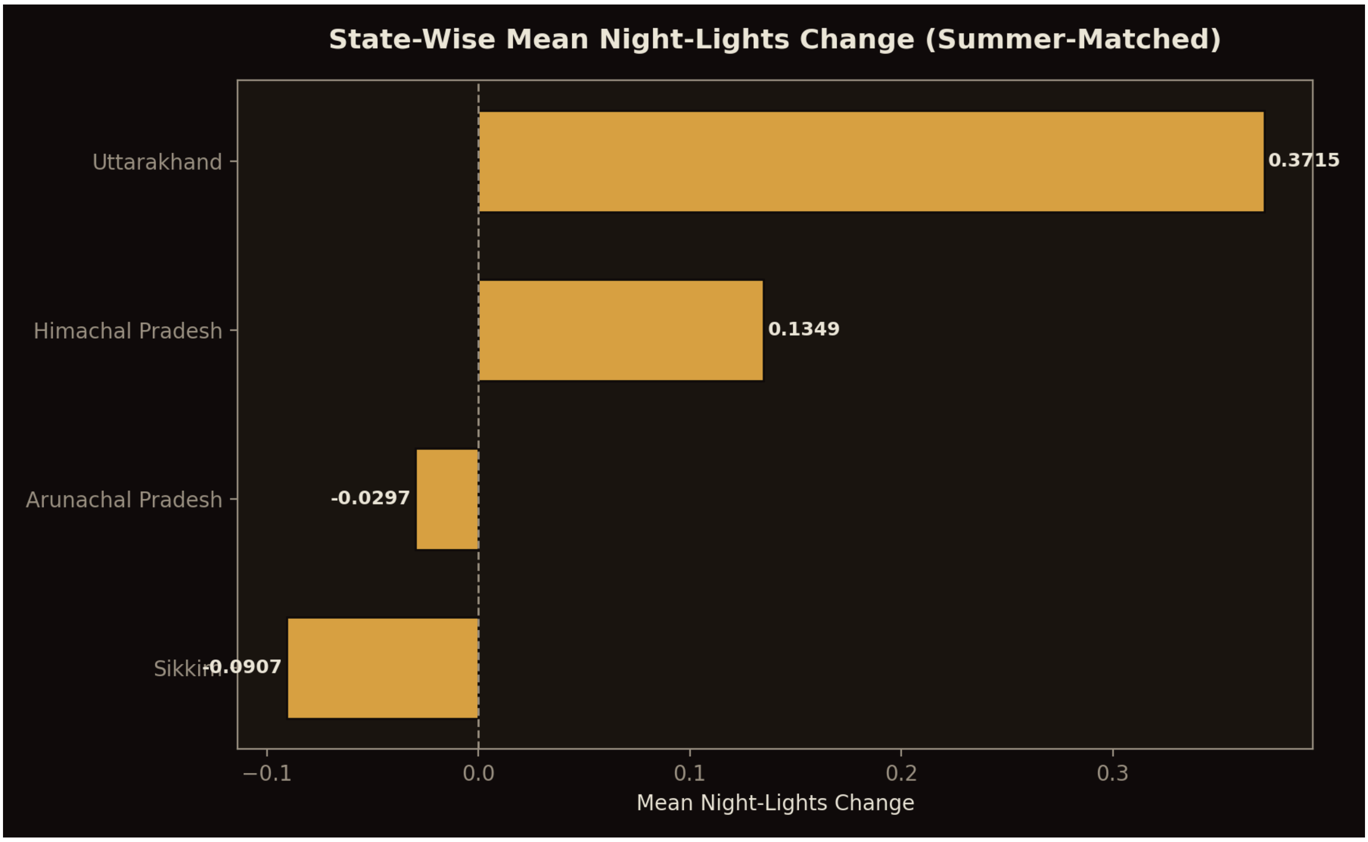

Figure 4. State-wise mean night-lights change (summer-matched composite), by state.

4.4 Budget Independence (RQ2)

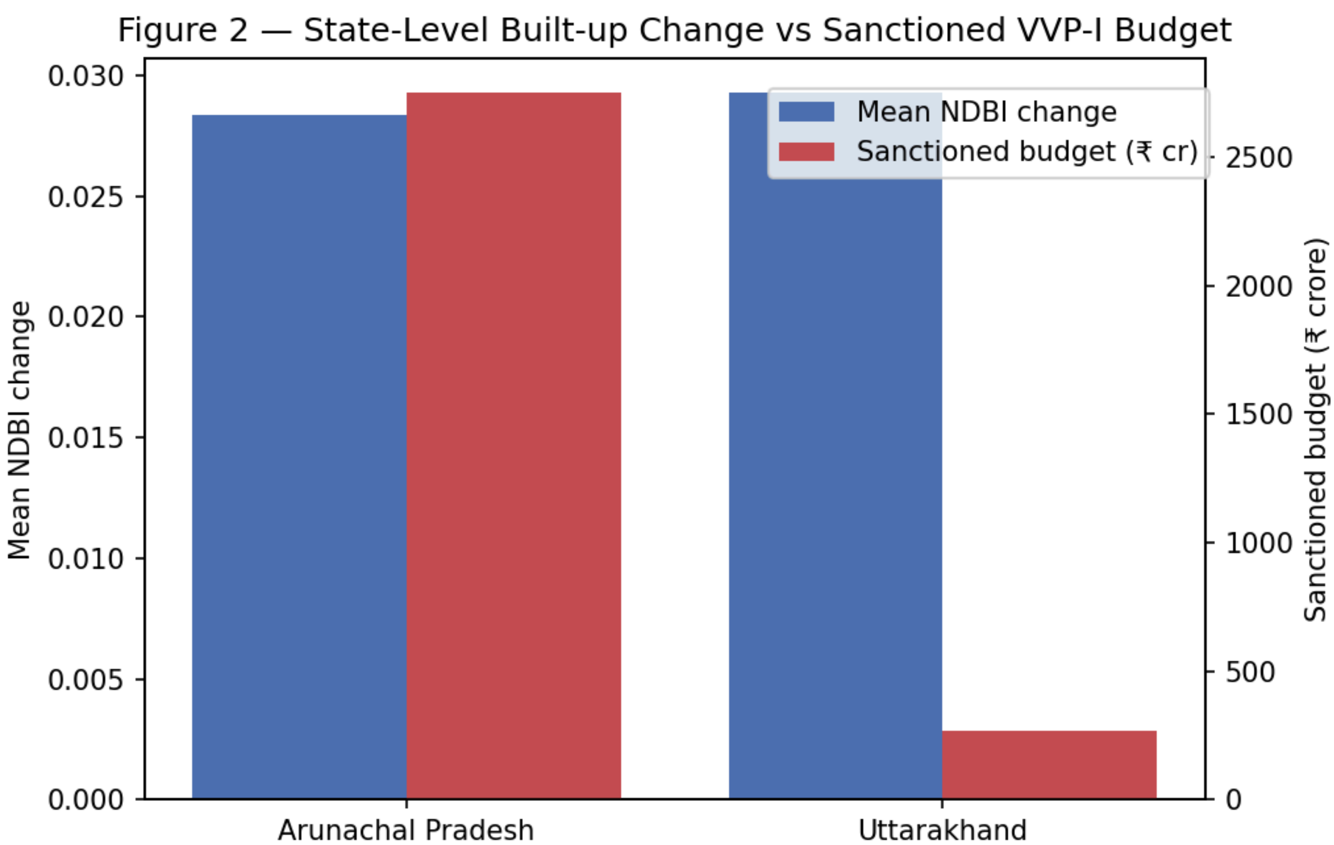

Figure 5. State-level mean built-up-area change plotted against sanctioned VVP-I budget.

Restricting to states with sufficient valid summer-window NDBI coverage left two comparable data points: Arunachal Pradesh (120 villages, mean NDBI change +0.0284; ₹2,749.74 crore sanctioned, 2,082 projects) and Uttarakhand (34 villages, mean NDBI change +0.0293; ₹270.58 crore sanctioned, 200 projects). Despite a roughly tenfold difference in sanctioned budget and project count, the two states' mean measured built-up change is nearly identical. Two data points cannot support a formal statistical claim, but this pattern is descriptively consistent with H2: budget scale is not translating proportionally into a correspondingly larger physical development signal.

4.5 Border-Proximity Testing (H3)

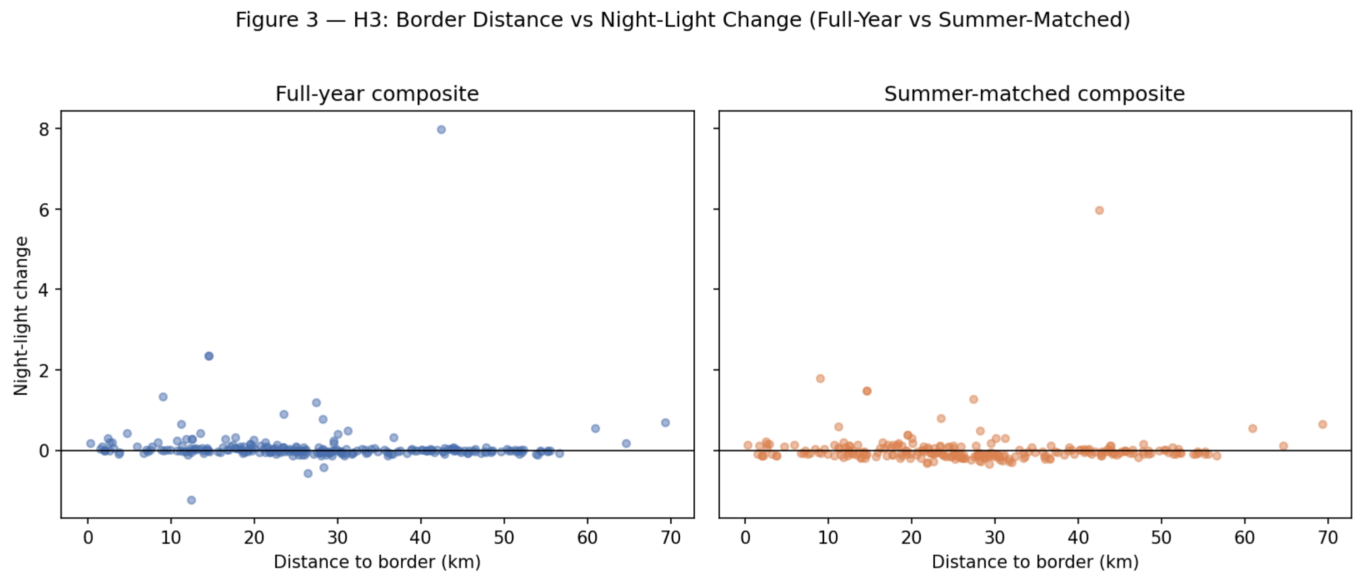

Figure 6. Distance-to-border versus NDBI change and night-lights change, both compositing windows.

Distance to the border/LAC ranged from 0.1 km to 69.4 km across the sample (mean 27.3 km). NDBI change showed no relationship with border proximity in either compositing window (full-year $\rho = 0.043$, $p = 0.497$; summer-matched $\rho = 0.037$, $p = 0.650$) — a stable non-finding. Night-lights change showed a significant relationship in the full-year window ($\rho = -0.259$, $p < 0.0001$, consistent with H3's prediction that closer villages change more) but not in the summer-matched window ($\rho = -0.076$, $p = 0.233$) — the same window-sensitivity pattern documented in Section 4.2, and reported with the same caution rather than as a confirmed result.

4.6 Robustness Summary

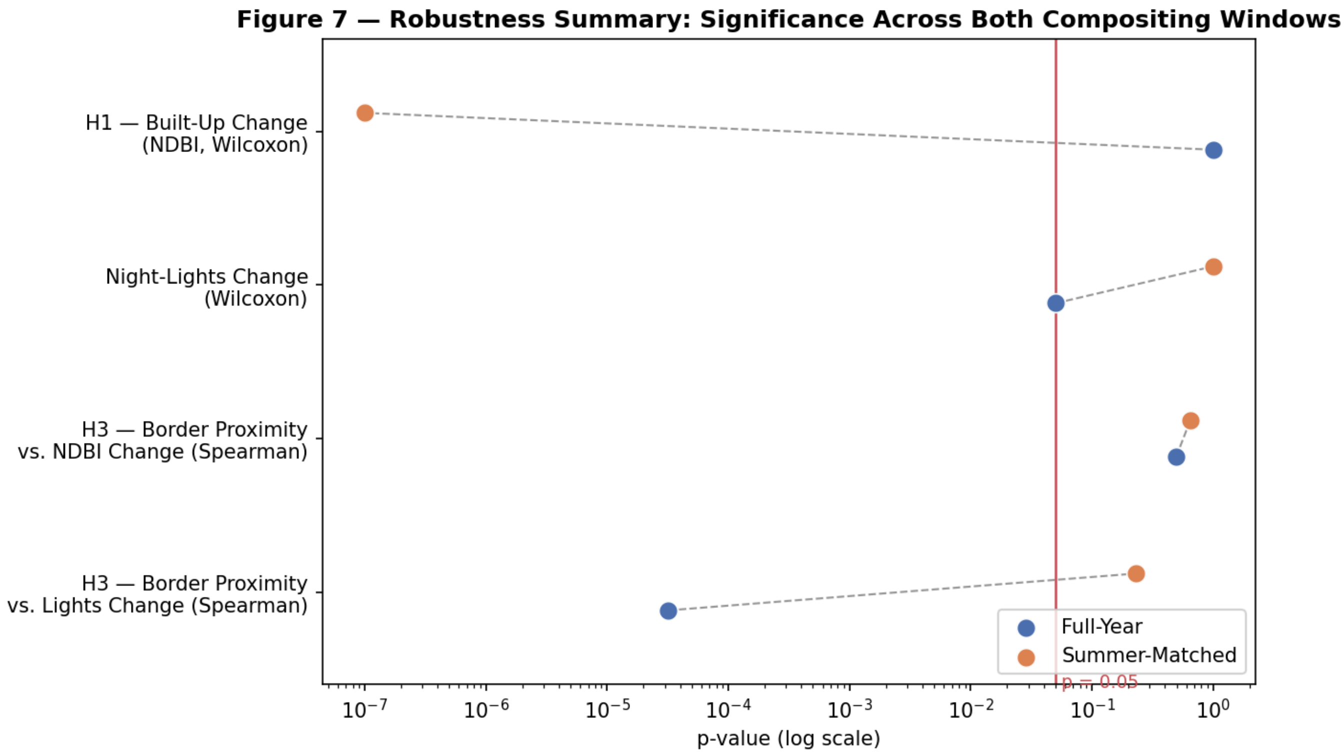

Figure 7. Significance (p-value, log scale) for all four tests under both compositing windows, with a reference line at $p = 0.05$. A test whose two points fall on opposite sides of the line reverses conclusion depending on which window is used.

Read together, the four tests split into two distinct patterns rather than one. H1 (built-up change) and the H3 night-lights correlation both cross the $p = 0.05$ line between windows — full-year non-significant, summer-matched significant for H1 (and the reverse for H3 night-lights) — the signature of a compositing-window-sensitive result. The H3 NDBI-proximity test, by contrast, stays non-significant in both windows ($p = 0.497$ full-year, $p = 0.650$ summer-matched): a genuinely stable null, not merely a result that happened not to flip. This distinction matters for how each finding should be read — the unstable results are reported as open questions, not as confirmed effects in either direction, while the stable null is reported with more confidence precisely because it did not depend on which window was chosen.

Across the eight tests reported here (four tests × two windows), no multiple-testing correction was applied to the headline p-values, since each test addresses a distinct research question rather than one hypothesis tested eight ways. As a conservative check, applying a Holm-Bonferroni correction across all eight simultaneously (family-wise $\alpha = 0.05$) leaves the paper's two significant results intact — H1 summer-matched NDBI ($p = 9.3 \times 10^{-15}$) and H3 full-year night-lights ($p = 3.2 \times 10^{-5}$) both clear even the strictest step in the correction — while every result already treated as non-significant in this paper remains non-significant. The correction does not change any conclusion; it confirms that the two significant results are not artifacts of running several tests.

5. Discussion

The central finding of this study is not a confirmed increase or decrease in physical development, but a demonstrated instability in the only metric that produced a "significant" result at all. A single-date, single-window spectral comparison in high-relief Himalayan terrain is not, on its own, sufficient grounds for a confident claim about built-up change — precisely because snow cover and monsoon cloud cover both vary by season in ways that can flip a result's sign independent of any true change on the ground. Night-lights, the more temporally stable proxy available, shows no confirmed increase under either window, and where budget and change can be directly compared, a tenfold difference in sanctioned investment did not correspond to a proportionally larger measured effect. Read together with the border-proximity results — a stable null for built-up change and an unstable, window-sensitive result for night-lights — this study's evidence does not support a confident claim that VVP-I's sanctioned investment has produced measurable, verifiable development at the pace or scale its budget would imply, nor that its distribution is confidently explained by border-proximity-driven prioritization over developmental need. This is precisely the accountability gap the programme's own parliamentary record identifies: a scheme of this scale, framed simultaneously as welfare and as strategic signal, has never been independently assessed, and this study's own results illustrate why that gap is consequential rather than resolving it with a convenient answer in either direction.

6. Limitations

6.1 Village Coverage Gaps

Ladakh (35 sanctioned villages) is fully excluded from village-level analysis; no publicly indexed source names its villages, and closing this gap would require an RTI request not pursued within this study's scope. Himachal Pradesh (7 of 51 inhabited priority villages identified) is carried only as an illustrative case study, not the core statistical sample. Nineteen of Uttarakhand's Pithoragarh villages have confirmed names but unresolved block assignment.

6.2 Border Geometry as Cartographic Approximation

Distance-to-border figures rely on Natural Earth's cartographic boundary line, a simplification of a disputed Line of Actual Control with no single internationally agreed alignment, and should be read as relative rather than authoritative.

6.3 Budget-Correlation Scope and Ecological Inference (RQ2)

The RQ2 budget-correlation observation rests on only two states with sufficient valid data and is reported descriptively rather than as a statistically confirmatory result; it also compares a state-level sanctioned budget against a village-level sampled mean, an ecological-inference gap — the budget figure describes spending across the whole state's VVP-I portfolio, not specifically the villages this study happened to geocode, so the comparison is suggestive rather than a clean like-for-like test.

6.4 Compositing-Window Sensitivity

Most centrally, the built-up-area and border-proximity findings are compositing-window-sensitive rather than stable, and this instability is reported as a limitation of single-date spectral comparison in this terrain, not resolved by preferring whichever result looks cleanest.

6.5 Geocoding Coverage and Selection Bias

Only 258 of 559 attempted villages (46%) were successfully geocoded, and the drop-off is uneven — Arunachal Pradesh alone lost 269 of 455 (Section 4.1). A village that fails to geocode against OpenStreetMap or Bhuvan could plausibly be smaller, more remote, or less administratively documented than one that succeeds, which would bias the analyzed sample toward already-more-accessible villages. This was tested directly rather than assumed away: for Arunachal Pradesh, where Census 2011 population and household counts are available for every village in the raw list regardless of geocoding outcome, a Mann-Whitney U test found no significant difference between geocoded and non-geocoded villages in either population (matched mean 135.0 vs. unmatched 145.7; p = 0.310) or households (matched mean 25.8 vs. unmatched 28.8; p = 0.540). Geocoding success in this dataset is not detectably associated with village size, which weighs against — though cannot fully rule out — a size-driven selection bias in the analyzed sample. Sikkim and Uttarakhand's raw village lists do not carry population or household fields, so this check could not be extended to them.

6.6 No Ground-Truth Validation

This study is satellite-only: no known-completed VVP-I project (e.g., one confirmed by a dated press release) was used as a positive control to verify that NDBI and VIIRS are sensitive enough, at 500m resolution, to detect development at the scale VVP-I typically funds (a village road, a school building, a handful of homes). A search for a specific, by-name, dated positive control during this study's review did not turn up a completed project that also appears in the 258-village geocoded sample, so this check could not be completed with real data rather than a placeholder; it is carried forward as a concrete item in Future Work rather than skipped silently.

7. Future Work

The following are concrete extensions identified during this study's own review process, not yet implemented, and deliberately kept separate from the Discussion and Limitations sections above so that what was actually done is never conflated with what remains open.

7.1 SAR-Based Change Detection

Sentinel-1 Synthetic Aperture Radar (VV/VH backscatter or coherence change) is not affected by cloud cover or the optical snow-contamination confound that motivated this study's compositing-window robustness check in the first place. Radar-based change detection would not resolve the debate between the full-year and summer-matched optical results — it would provide a genuinely independent third measurement, immune to both of this study's identified confounds, which could help determine whether either optical result is closer to the truth.

7.2 Building-Footprint or Sub-500m Structural Analysis

NDBI at a 500m buffer averages a mix of built-up surface, bare rock, agricultural land, and natural vegetation in a single value — a coarse proxy for what VVP-I actually funds at the village level (individual roads, buildings, small installations). High-resolution optical imagery (PlanetScope) or open building-footprint datasets (e.g., Google/Microsoft building footprints) could support structure-level change detection instead of an area-averaged index.

7.3 A Genuine Control-Group (Difference-in-Differences) Design

This study compares each village's own before/after values; it does not compare VVP-I priority villages against a matched set of non-VVP border villages experiencing the same regional trends over the same period. Adding such a control group would let a Difference-in-Differences design isolate a VVP-I-attributable effect from a general regional development trend, rather than reporting an uncontrolled before/after change as this study does.

7.4 Multi-Year, Phenology-Normalized Trend

This study compares two single years (2021, 2025); a three-or-more-point trend (e.g., 2021, 2023, 2025) using multi-year cloud-free medians rather than single-year composites would reduce sensitivity to any one year's anomalous weather and let a trend, rather than a two-point difference, carry the evidentiary weight.

7.5 Buffer-Size Sensitivity Testing

The 500m buffer radius was fixed rather than tested; a sensitivity sweep (e.g., 250m, 500m, 1km) run with the same compositing-window-robustness discipline already applied elsewhere in this study would establish whether the reported results are stable to this choice or, like the compositing window, sensitive to it.

7.6 Ground-Truth Positive-Control Validation

As noted in Limitations, no by-name, dated, independently-confirmed completed VVP-I project was identified during this study that also falls within the 258-village geocoded sample. Identifying even two or three such villages — for instance from a dated press release or news report confirming a specific completed road or building — and checking whether their individual NDBI/VIIRS signal is detectable would directly test this study's implicit assumption that the chosen indices and resolution are sensitive enough to detect the scale of development VVP-I typically funds.

7.7 RTI Follow-Through for Himachal Pradesh and Ladakh

Development_Log.md records a deliberate decision not to file Right to Information requests for the two states' missing village-wise annexures, given this study's timeline. Filing those requests remains open and would close the two largest documented coverage gaps with primary-source data rather than the illustrative/excluded treatment used here.

8. Conclusion

This study finds that India's Vibrant Villages Programme (VVP-I), sanctioned at a scale of approximately ₹4,800 crore across five border states/union territories, cannot currently be shown, with confidence, to have produced measurable, verifiable physical development at its priority villages — not because no change is detectable, but because the one metric that shows a significant increase does so in only one of two equally defensible measurement windows, while the more temporally stable available proxy shows no increase under either. Where budget scale can be directly compared against measured change, a tenfold difference in investment did not correspond to a proportional difference in outcome. The resulting implication is direct: a scheme of VVP-I's scale and strategic framing should not continue without the independent impact assessment its own parliamentary record confirms has never been conducted — and any future assessment should be held to the same compositing-window robustness standard applied here, rather than reporting whichever single-window result is most convenient.

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